

HYEONBEEN LEE

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EDUCATION

Virginia Tech, Blacksburg, VA, United States Aug. 2026 — May 2031 (Expected)

Ph.D. in Mechanical Engineering (Advisor: [Simon Stepputtis](#))

Research Focus: Vision-Language-Action (VLA), Contact-rich manipulation, Real-time controller

Kyung Hee University, Yongin, South Korea

Mar. 2022 — Feb. 2024

M.Eng. in Mechanical Engineering (Advisor: [Jin-Gyun Kim](#))

GPA: 3.93/4.0

Thesis: ‘Composite neural network with differential propagation for modeling impulsive nonlinear dynamic systems’

Kyung Hee University, Yongin, South Korea

Mar. 2015 — Feb. 2022

B.Eng. in Mechanical Engineering (Advisor: [Jin-Gyun Kim](#), [Shinkyu Jeong](#))

(Upper-Division: 3.88/4.0) GPA: 3.52/4.0

Thesis: ‘Data-driven aerodynamic coefficient prediction using deep neural network and PARSEC airfoil parameterization’

(* Military leave of absence: May 2017 — Feb. 2019)

PUBLICATIONS

Journal Papers

4. **H. Lee**, M. Jung, T.K. Yeu, J.B. Han, D. Park, S. Stepputtis, J.G. Kim (2026), ‘Frequency-aware decomposition learning for sensorless wrench estimation in vibration-rich robotic contact’, submitted to *Robotics and Computer-Integrated Manufacturing*. doi:10.48550/arXiv.2604.12905 [\[preprint\]](#) [\[code\]](#) [\[dataset\]](#)
3. B. Koo, M. Jung, **H. Lee**, T. Yeu, J.G. Kim, J.B. Han, Y. Lee, D. Park (2026), ‘Buoyancy-integrated hybrid reaction force estimation method with real-time haptic feedback for underwater hydraulic manipulation’, revised from *IEEE Robotics and Automation Letters*.
2. **H. Lee**, S. Han, H.S. Choi, J.G. Kim (2024), ‘cNN-DP: Composite neural network with differential propagation for impulsive nonlinear dynamics’, *Journal of Computational Physics (JCR IF Top 2.5% in Physics, Mathematical)*, 496, 112578. doi: 10.1016/j.jcp.2023.112578 [\[paper\]](#) [\[code\]](#)
1. S. Han*, G.E. Jeong*, **H. Lee**, W.S. Choi, J.G. Kim (2023), ‘Multi-body dynamics model for spent nuclear fuel transportation system under normal transport test conditions’, *Nuclear Engineering and Technology (JCR IF Top 13.7% in Nuclear Science & Technology)*, 55(11), 4125-4133. (*Co-first authors) doi: 10.1016/j.net.2023.07.033 [\[paper\]](#)

Conference Proceedings & Presentations

6. J.H. Han, J.B. Han, S.S. Kim, M.H. Kim, Y.H. Kim, **H. Lee**, J.G. Kim, T.K. Yeu, ‘Digital twin model of underwater construction robot for real-time grinding simulation’, *7th International Conference on Multibody System Dynamics (IMSD)*, Madison, WI, United States, Jun. 2024. [\[abstract\]](#)
5. **H. Lee**, J. Han, T.K. Yeu, J.G. Kim, ‘Real-time multi-horizon reaction force forecasting of marine robot using interpretable Transformer’, *Korean Society of Mechanical Engineers (KSME)* (Oral Presentation), Incheon, South Korea, Nov. 2023. [\[abstract\]](#)
4. **H. Lee**, S. Han, H.S. Choi, J.G. Kim, ‘Meta-modeling of nonlinear impulsive dynamics using composite neural network model with differential propagation’, *Korean Society of Mechanical Engineers (KSME)* (**Excellent Paper Award**, Oral Presentation, * Extended works from IMAC 2023), Busan, South Korea, May 2023. [\[abstract\]](#)
3. **H. Lee**, S. Han, H.S. Choi, J.G. Kim, ‘Composite neural network with differentiation propagation for modeling nonlinear impulsive dynamics’, *Conference on Dynamics and Control in Korean Society of Mechanical Engineers (KSME)* (Oral Presentation), Jeju, South Korea, Mar 2023. [\[abstract\]](#)
2. **H. Lee**, S. Han, H.S. Choi, J.G. Kim, ‘Composite neural network framework for modeling impulsive nonlinear dynamic responses’, *41st International Modal Analysis Conference (IMAC-XLI)* (Oral Presentation), Austin, TX, United States, Feb. 2023. doi: 10.1007/978-3-031-34946-1.21 [\[paper\]](#)
1. **H. Lee**, J.G. Kim. ‘Development of multibody dynamics trailer model using normal transportation test data and DNN based surrogate model generation’, *Korean Society for Noise and Vibration Engineering (KSNVE)* (Oral Presentation), Jeju, South Korea, Dec 2022.

AWARDS & HONORS

- **Excellent Paper Award** Korean Society of Mechanical Engineers, 2023
 - Awarded for a first-author presentation at KSME 2023 conference.
- **Brain Korea 21 Plus Scholarship** Graduate School, Kyung Hee University, 2022—2023
 - Received \$18,000 during M.Eng. studies through a national graduate research scholarship.
- **Graduate Assistantship** Dept. of Mechanical Engineering, Kyung Hee University, 2022—2023
 - Received \$12,000 during M.Eng. studies through a selective departmental assistantship.
- **Academic Superiority Scholarship** Dept. of Mechanical Engineering, Kyung Hee University, 2021
 - Ranked 1st in the departmental merit review and received a full tuition waiver.

RESEARCH EXPERIENCES

Virginia Tech, Department of Mechanical Engineering

Thinking Embodied Agents Lab., advised by *Dr. Simon Stepputtis*

Graduate Research Assistant

Aug. 2026—Present

- Developing hierarchical Vision-Language-Action (VLA) systems and sensorless contact perception methods for real-time robot controllers.

Kyung Hee University, Department of Mechanical Engineering

Modeling & Simulation Lab., advised by *Dr. Jin-Gyun Kim*

Research Assistant

Mar. 2024—Apr. 2026

- Proposed a frequency-aware decomposition learning framework for sensorless wrench estimation in contact- and vibration-rich robotic contact. Submitted a first-authored manuscript.
- Optimized the developed wrench estimation model architectures for real-time closed-loop deployment with Korea Research Institute of Ships and Ocean Engineering (KRISO).

KAIST, Kim Jaechul Graduate School of Artificial Intelligence

Cognitive Learning for Vision & Robotics Lab., advised by *Dr. Joseph J. Lim*

Research Intern

Sep. 2023—Nov. 2023

- Investigated skill-prior regularization method for accelerating reinforcement learning in long-horizon robotic manipulation.

Kyung Hee University, Department of Mechanical Engineering

Modeling & Simulation Lab., advised by *Dr. Jin-Gyun Kim*

Graduate Research Assistant

Mar. 2022—Feb. 2024

- Proposed a composite neural network architecture for learning impulsive high-order dynamics from multi-order derivatives. Published a first-author article in the *Journal of Computational Physics*.
- Developed a multibody dynamics vehicle model for the Korea Atomic Energy Research Institute (KAERI). Published a co-authored article in the *Nuclear Engineering and Technology*.
- Designed a data-driven model for sensorless wrench estimation in a 6-DoF hydraulic robot grinding for the Korea Research Institute of Ships and Ocean Engineering (KRISO).

Kyung Hee University, Department of Mechanical Engineering

Modeling & Simulation Lab., advised by *Dr. Jin-Gyun Kim*

Undergraduate Research Assistant

Jan. 2021—Feb. 2022

Aerodynamic Design Exploration & Big Data Analysis Lab, advised by *Dr. Shinkyu Jeong*

Sep. 2020—Jun. 2021

- Investigated data-driven methods for flexible multibody dynamics and aerodynamic design optimization and led an undergraduate thesis project.

INVITED SEMINARS

2. “On the data-driven modeling of contact- and vibration-rich dynamics”, invited seminar, IMBD Lab, Chungnam National University, Daejeon, South Korea, Apr. 16, 2026 [\[keynote\]](#)
1. “On the data-driven modeling of contact- and vibration-rich dynamics”, invited seminar, Modeling & Simulation Lab, Kyung Hee University, Yongin, South Korea, Apr. 14, 2026 [\[keynote\]](#)

TEACHING & MENTORSHIP

Research Mentor

— Guided junior graduate students on data-driven modeling and robotics.

Modeling & Simulation Lab., Kyung Hee University

Mar. 2025—Jul. 2026

Teaching Assistant, ME380 Systems Dynamics

— Lectured on multibody dynamics with theoretical derivation and its computational implementation.

Dept. of Mechanical Engineering, Kyung Hee University

Mar. 2022—Jun. 2023

COMMUNITY SERVICE

Lead Administrative Assistant
48th College of Engineering Student Council
ROK-U.S. Combined Marine Corps Interpreter

Dept. of Mechanical Engineering, Kyung Hee University, 2022—2024
Kyung Hee University, Feb 2019—Jan. 2020
1st Marine Div., ROKMC, Sep 2017—Feb. 2019

REFERENCES

- **Jin-Gyun Kim**, Ph.D.

Associate Professor at Department of Mechanical Engineering, Kyung Hee University (jingyun.kim@khu.ac.kr)

- **Hee-Sun Choi**, Ph.D.

Assistant Professor at Department of Artificial Intelligence and Robotics, Sejong University (heesunchoi@sejong.ac.kr)

- **Seongji Han**, Ph.D.

Assistant Professor at Department of Mechatronics Engineering, Chungnam National University (seongji.han@cnu.ac.kr)